A Novel Distributed Bipartite Consensus Control of Nonlinear Multiagent Systems via Prioritized Strategy Approach

Xinjun Wang, Ben Niu, Lun Zhai, Jie Kong, and Xiaomei Wang

***Abstract*—In this brief, the adaptive bipartite consensus tracking problem is considered for nonlinear MASs over signed directed graphs with unknown nonlinear functions in high-order and nonstrict-feedback form. By proposing a prioritized strategy, a more relaxation condition that a signed digraph only containing a spanning tree is used in this brief, and all the agents in the system can be achieved a bipartition. Besides, it is worth pointing out that the higher-order form and the nonstrict-feedback structure are a part of the each follower model, which makes the controller design process more complicated. A distributed neural-based adaptive backstepping technology is applied, where the unknown nonlinear functions are approximated by Neural networks (NNs). Under the proposed protocol, all the agents achieve the bipartite consensus with bounded errors. The feasibility of the newly designed method are verified by MATLAB simulation analysis.**

***Index Terms*—Multi-agent systems, bipartite consensus, adaptive backstepping, neural networks.**

I. Introduction

THE RESEARCH on cooperative consensus of nonlinear multi-agent systems (MASs) has become the current hot topics because of its countless applications in multi-spacecraft systems, multi-robot pursuit-evasion, and manufacturing systems, for instance, [1]–[3] and the references therein. Given their widespread applications, consensus problem has been received considerable attention and a variety of results over nonnegative graphs have been explored in [4]–[6]. However, antagonistic interactions also exist in numerous situations, such as competition robotic soccer [7], social networks [8], opinion dynamics [9], where the communications are denoted via signed graphs. In other words, there exist edges with positive and negative weights, simultaneously.

Manuscript received November 30, 2021; accepted December 7, 2021. Date of publication December 10, 2021; date of current version May 27, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61873151; in part by the Shandong Provincial Natural Science Foundation of China under Grant ZR2019MF009; and in part by the Taishan Scholar Project of Shandong Province of China under Grant tsqn201909078. This brief was recommended by Associate Editor I. W. H. Ho. *(Corresponding author: Ben Niu.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TCSII.2021.3134253.

Digital Object Identifier 10.1109/TCSII.2021.3134253

Recently, the approaches of bipartite consensus tracking containing both leaderless and the lead-follower of MASs over signed graphs were provided in [8]. All agents in [8] can achieve bipartite consensus by proposing Laplacian feedback design methods. Moreover, when considering a class of general linear MASs, the bipartite consensus control method was pondered in [10], [11]. It is notable that the above results are only considered the linear integrator dynamics. However, in many actual applications, MASs are inherently nonlinear and uncertain dynamics [12], [13]. Thus, numerous researchers have devoted themselves to the study of bipartite consensus control design for nonlinear MASs [14]–[16]. Nevertheless, most of actual models do not satisfy nonlower-triangular structures [14], [15], and the structurally balanced condition needs to be satisfied.

On the other hand, the aforementioned works require that the considered virtual item and real control input item of MASs satisfy the linear form, which cannot satisfy many physical system that are governed by the high-order format [17]. In order to study and tackle the control problem of high-order nonlinear systems, a novel adaptive control technique focusing on the issue of adding a power integrator was proposed in [18]. Then, using the above method, many remarkable results have been obtained in the research of a wide range of high-order systems [19]–[22]. However, all the aforementioned results only focus on the single systems, while a few results have been investigated for high-order MASs.

This brief aims to present bipartite consensus tracking control strategy for nonlinear MASs in high-order and nonstrict-feedback form. The novelties of this brief are marked as follows: (1) Most of the current bipartite consensus works [23], [24] only concentrate upon nonlinear MASs with a signed diagraph to be structurally balanced, and there are few significant results in bipartite consensus control for nonlinear MASs under a signed diagraph to be structurally unbalanced. A more relaxed condition that a signed digraph only containing a spanning tree is used in this brief, while the strict limitation to a signed digraph be structurally balanced is removed. By utilizing the idea of a shortest path, a prioritized strategy is introduced to achieve a bipartition of the agents in this brief. On this basis, the bipartite consensus is successfully achieved under the designed control protocol for the considered high-order nonlinear MASs. (2) The proposed distributed neural-based adaptive scheme proves the boundedness of all the closed loop signals. It is worth pointing out

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that the non-strict feedback form is also successfully handled by applying a property of NNs. Thus, the proposed result is more general than the ones being in lower-triangular structures [23], [24].

II. Problem Formulation and Preliminaries

A. Graph Theory

Consider $G \triangleq \{O, E, A\}$ represent a weighted signed digraph of order N, $O = \{o_i : 1, \ldots, N\}$ represents a set of follower agents, $E \subseteq \{e_{ij} : i = 1, \ldots, N, j = 1, \ldots, N, i \neq j\}$ denotes a set of edges, and adjacency matrix is described as $A = [a_{ij}] \in R^{N\times N}$, where $a_{ij} < 0$ means the antagonistic interaction and $a_{ij} > 0$ the cooperative interaction; otherwise $a_{ij} = 0$. The set of neighbors of ith follower is $N_i = \{j \mid a_{ij} \neq 0\}$. $L = D - A$ is a Laplacian matrix, where $L \in R^{N\times N}$, $D = \mathrm{diag}(d_i)$ is the in-degree matrix of ith follower, and $d_i = \sum_{j\in N_i} |a_{ij}|$.

Remark 1: The readers may refer to [5] for more details about Graph theory.

Assumption 1: $\bar{G}$ contains a directed spanning tree, and the leader is selected as the root node, only the case with one leader is considered.

Definition 1: If the agents $o_1, o_p \in E$, a shortest path $P_1 : o_p \to o_1$ (o_1 can obtain the information from o_p) is defined:

$$P_1 = \min\big\{(o_1, o_2), (o_2, o_3), \ldots, (o_{p-1}, o_p)\big\} \subset E \tag{1}$$

A path $P_1 > 0$ if the edge weights $a_{1,2} \cdot a_{2,3} \cdots a_{p-1,p} > 0$; $P_1 < 0$ if $a_{1,2} \cdot a_{2,3} \cdots a_{p-1,p} < 0$. S_{P_1} is defined

$$S_{P_1} = \begin{cases} 1, & P_1 > 0 \\ -1 & P_1 < 0 \end{cases} \tag{2}$$

and $L_{P_1} = p - 1$ denotes length of P_1.

B. Prioritized Strategy

In order to achieve a bipartition for a network satisfying Assumption 1, a prioritized strategy will be described according to Definition 1. For example, for an agent o_i, the information can be received from the leader o_0 by selecting different paths, and from these different paths, there exists at least a shortest information transmission path $P_{i,1}$. On the other hand, we define a set $\Omega_0 = \{o_0\}$ for the leader o_0 and a set $\Omega_1 = \{o_i | L_{P_i} = 1\}$ for those agents directly obtaining the information from the leader v_0. Define $\Omega_2 = \{v_i | L_{P_i} = 2\}$, $\ldots$, $\Omega_l = \{o_i | L_{P_i} = l\}$ until all agents are included.

In a shortest route, if the agents $o_i \in \Omega_k$, $o_j \in \Omega_{k-1}$, ($1 \leq k \leq l$), an unique agent o_j for agent o_i can be chosen as $j = \min\{\max\{|a_{i,j}|, o_i \in \Omega_k, o_j \in \Omega_{k-1}\}\}$, if $\max\{|a_{i,j_1}| = |a_{i,j_2}|, o_i \in \Omega_k, o_{j_1}, o_{j_2} \in \Omega_{k-1}\}$, then an unique agent o_j for o_i can be chosen as $j = \min\{j_1, j_2\}$.

C. System Description

The ith follower are described as follows:

$$\begin{aligned} \dot{x}_{i,m'} &= g_{i,m'}(\bar{x}_{i,m'})x_{i,m'+1}^{p_{i,m'}} + f_{i,m'}(\bar{x}_i) \\ \dot{x}_{i,n_i} &= g_{i,n_i}(\bar{x}_{i,n_i})u_i^{p_{i,n_i}} + f_{i,n_i}(\bar{x}_i) \\ y_i &= x_{i,1} \end{aligned} \tag{3}$$

where $m' = 1, \ldots, n_i - 1$; $\bar{x}_{i,m} = [x_{i,1}, x_{i,2}, \ldots, x_{i,m}]^T$; ($i = 1, \ldots, N, m = 1, \ldots, n_i$); $\bar{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,n_i}]^T \in R^{n_i}$, ($i = 1, \ldots, N$) stands for the system state of the ith follower, $y_i \in R$ is the system output. $g_{i,m}(\bar{x}_{i,m})$ and $f_{i,m}(\bar{x}_i)$ ($i = 1, \ldots, N, m = 1, \ldots, n_i$) are smooth functions and $f_{i,m}(\bar{x}_i)$ is unknown. For $i = 1, \ldots, N, m = 1, \ldots, n_i$, $g_{i,m}$ is known, there exist constants $g_{\min}$ and $g_{\max}$ such that $0 < g_{\min} \leq g_{i,m} \leq g_{\max}$ and $|\dot{g}_{i,m}| \leq g_d$ with $g_d > 0$. $u_i \in R$ is the protocol and $p_{i,m}$ are positive odd integers.

D. Neural Networks

Lemma 1 [25]: Let $\bar{\chi}_p = [\chi_1, \chi_2, \ldots, \chi_p]^T$ and $S(\bar{\chi}_p) = [s_1(\bar{\chi}_p), s_2(\bar{\chi}_p), \ldots, s_l(\bar{\chi}_p)]^T$ represents basis function of NNs, if q and p are positive integers ($q \leq p$), it has the following property:

$$\big\|S(\bar{\chi}_q)\big\|^2 \geq \big\|S(\bar{\chi}_p)\big\|^2 \tag{4}$$

Lemma 2 [26]: If x and y are real-valued functions, the following results hold:

$$\big|x^p - y^p\big| \leq p\big|x - y\big|\big|x^{p-1} - y^{p-1}\big| \tag{5}$$

$$\big|x + y\big|^c \leq 2^{c-1}\big(\big|x^c\big| + \big|y^c\big|\big) \tag{6}$$

where $p \in R^* \triangleq \{q \in R^+ : q \geq 1$ and q is an odd integer$\}$, $c \geq 1$.

III. Main Results

In this part, the distributed bipartite consensus protocols are developed for MASs (3) based on backstepping technique and prioritized strategy. First, for MAS (3), we define $p_i = \max\{p_{i,l}, l = 1, 2, \ldots, n_i\}$. Moreover, the unknown constants are defined as

$$\theta_{i,k} = \big\|\Phi_{i,k}\big\|^{\frac{p_i+1}{p_i - p_{i,k}+1}}, i = 1, \ldots, N, k = 1, \ldots, n_i \tag{7}$$

Let use $\hat{\theta}_{i,k}$ express the estimation of $\theta_{i,k}$. Thus, $\tilde{\theta}_{i,k} = \theta_{i,k} - \hat{\theta}_{i,k}$ denotes the estimated error.

Consider the agent o_i ($i \in \Omega_k, k \geq 1$) with its unique agent o_j ($j = \min\{\max\{|a_{i,j}|, v_j \in \Omega_{k-1}\}\}$) in path P_i, the following bipartite consensus tracking errors are defined as

$$\begin{aligned} e_{i,1} &= \big(x_{i,1} - \mathrm{sig}\big(a_{ij}\big)x_{j,1}\big) + \big(x_{i,1} - S_{P_i}y_0\big) \\ e_{i,k} &= x_{i,k} - \alpha_{i,k-1}, \quad k = 2, \ldots, n_i \end{aligned} \tag{8}$$

where $\mathrm{sig}(\cdot)$ is the standard sign function, $\alpha_{i,k-1}$ is virtual control protocol.

Step 1: For $k = 1$, define

$$V_{i,1} = \frac{e_{i,1}^{p_i - p_{i,1}+2}}{p_i - p_{i,1} + 2} + \frac{1}{2l_{i,1}}\tilde{\theta}_{i,1}^2 \tag{9}$$

where $l_{i,1} > 0$ is a design constant.

Differentiating $V_{i,1}$ yields

$$\dot{V}_{i,1} = e_{i,1}^{p_i - p_{i,1}+1}\Big(2g_{i,1}x_{i,2}^{p_{i,1}} + \bar{f}_{i,1}\big(\chi_{i,1}\big)\Big) - \tfrac{1}{\tau_{i,1}}\tilde{\theta}_{i,1}\dot{\hat{\theta}}_{i,1} \tag{10}$$

where $\bar{f}_{i,1}(\chi_{i,1}) = 2f_{i,1}(\bar{x}_i) - \mathrm{sig}(a_{ij})(g_{j,1}x_{j,2} + f_{j,1}(\bar{x}_j)) - S_{P_i}\dot{y}_0$ with $\chi_{i,1} = [\bar{x}_i, \bar{x}_j, y_0, \dot{y}_0]^T \in \Omega$. The NNs is applied to estimate $\bar{f}_{i,1}(\chi_{i,1})$, i.e., $\bar{f}_{i,1}(\chi_{i,1}) = \Phi_{i,1}^T S_{i,1}(\chi_{i,1}) + \delta_{i,1}$, $\delta_{i,1}$ is the approximation error, $|\delta_{i,1}| \leq \xi_{i,1}$. According to

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Lemma 1, one has $\bar{f}_{i,1}(\chi_{i,1}) \leq \Phi_{i,1}^T S_{i,1}(\chi'_{i,1}) + \delta_{i,1}$, where $\chi_{i,1'} = [x_{i,1}, x_{j,1}, y_0, \dot{y}_0]^T$. Furthermore, using the Young's inequality, one has

$$\begin{aligned} e_{i,1}^{p_i-p_{i,1}+1}\bar{f}_{i,1}(\chi_{i,1}) &= e_{i,1}^{p_i-p_{i,1}+1}\left(\Phi_{i,1}^T S_{i,1}(\chi_{i,1}) + \delta_{i,1}\right) \\ &\leq \frac{p_i-p_{i,1}+1}{p_i+1}\gamma_1^{\frac{p_i+1}{p-p_{i,1}+1}} e_{i,1}^{p_i+1}\left\|\Phi_{i,1}\right\|^{\frac{p_i+1}{p_i-p_{i,1}+1}} \\ &\times \left(S_{i,1}^T S_{i,1}\right)^{\frac{p_i+1}{2(p_i-p_{i,1}+1)}} + \frac{p_{i,1}}{p_i+1}\gamma_{i,1}^{-\frac{p_i+1}{p_{i,1}}} \\ &+ \frac{p_i-p_{i,1}+1}{p_i+1}\eta_{i,1}^{\frac{p_i+1}{p_i-p_{i,1}+1}} e_{i,1}^{p_i+1} + \frac{p_{i,1}}{p_i+1}\eta_{i,1}^{-\frac{p_i+1}{p_{i,1}}}\xi_1^{\frac{p_i+1}{p_{i,1}}} \\ &\leq e_{i,1}^{p_i+1}\gamma_{i,1}^{\frac{p_i+1}{p_i-p_{i,1}+1}}\theta_{i,1}\left(S_{i,1}^T S_{i,1}\right)^{\frac{p_i+1}{2(p_i-p_{i,1}+1)}} + e_{i,1}^{p_i+1}\eta_{i,1}^{\frac{p_i+1}{p_i-p_{i,1}+1}} \\ &+ \gamma_{i,1}^{-\frac{p_i+1}{p_{i,1}}} + \eta_{i,1}^{-\frac{p_i+1}{p_{i,1}}}\xi_{i,1}^{\frac{p_i+1}{p_{i,1}}} \end{aligned} \tag{11}$$

where $\gamma_{i,1}$ and $\eta_{i,1}$ are positive design parameters.

Then, the virtual controller is constructed as

$$\alpha_{i,1} = -e_{i,1}\pi_{i,1} \tag{12}$$

where $\pi_{i,1} = \{\frac{1}{2g_{\min}}(c_{i,1} + \gamma_{i,1}^{\frac{p_i+1}{p_i-p_{i,1}+1}}\hat{\theta}_{i,1}(S_{i,1}^T S_{i,1})^{\frac{p_i+1}{2(p_i-p_{i,1}+1)}} + \eta_{i,1}^{\frac{p_i+1}{p_i-p_{i,1}+1}})\}^{\frac{1}{p_{i,1}}}$, $c_{i,1} > 0$ is the design parrameter. We chose adaptive law as

$$\dot{\hat{\theta}}_{i,1} = l_{i,1}\gamma_{i,1}^{\frac{p_i+1}{p_i-p_{i,1}+1}}\left(S_{i,1}^T S_{i,1}\right)^{\frac{p_i+1}{2(p_i-p_{i,1}+1)}} e_{i,1}^{p_i+1} - \mu_{i,1}\hat{\theta}_{i,1} \tag{13}$$

where $l_{i,1}$ and $\mu_{i,1}$ are the positive design parameters.

By applying (11), (12) and (13) to (10)

$$\begin{aligned} \dot{V}_{i,1} \leq& -c_{i,1}e_{i,1}^{p_i+1} + 2g_{i,1}e_{i,1}^{p_i-p_{i,1}+1}\left(x_{i,2}^{p_{i,1}} - \alpha_{i,1}^{p_{i,1}}\right) - \frac{\mu_{i,1}}{2\tau_{i,1}}\tilde{\theta}_{i,1}^2 \\ &+ \gamma_{i,1}^{-\frac{p_i+1}{p_{i,1}}} + \eta_{i,1}^{-\frac{p_i+1}{p_{i,1}}}\xi_{i,1}^{\frac{p_i+1}{p_{i,1}}} + \frac{\mu_{i,1}}{2\tau_{i,1}}\theta_{i,1}^2 \end{aligned} \tag{14}$$

Then

$$\begin{aligned} g_{i,1}e_{i,1}^{p_i-p_{i,1}+1}\left(x_{i,2}^{p_{i,1}} - \alpha_{i,1}^{p_{i,1}}\right) &\leq g_{\max}\left|e_{i,1}^{p_i-p_{i,1}+1}\right|\left|x_{i,2}^{p_{i,1}} - \alpha_{i,1}^{p_{i,1}}\right| \\ &\leq p_{i,1}g_{\max}\left|e_{i,1}^{p_i-p_{i,1}+1}\right|\left|e_{i,2}\right| \\ &\times \left|\left(e_{i,2}+\alpha_{i,1}\right)^{p_{i,1}-1} + \left(e_{i,1}\pi_{i,1}\right)^{p_{i,1}-1}\right| \\ &\leq p_{i,1}g_{\max}\left|e_{i,1}^{p_i-p_{i,1}+1}\right|\left|e_{i,2}\right| \\ &\times \left|2^{p_{i,1}-2}\left(e_{i,2}^{p_{i,1}-1} + \left(e_{i,1}\pi_{i,1}\right)^{p_{i,1}-1}\right) + \left(e_{i,1}\pi_{i,1}\right)^{p_{i,1}-1}\right| \\ &\leq e_{i,1}^{p_i+1} + \frac{1}{2}e_{i,2}^{p_i-p_{i,2}+1}\bar{\pi}_{i,1} \end{aligned} \tag{15}$$

where $\bar{\pi}_{i,1} = 2e_{i,2}^{p_{i,2}}((2^{p_{i,1}-2}g_{\max}p_{i,1})^{p_i+1/p_{i,1}} + (2^{p_{i,1}-2}p_{i,1}\pi_{i,1}^{p_{i,1}-1})^{p_i+1} + (g_{\max}p_{i,1}\pi_{i,1}^{p_{i,1}-1})^{p_i+1})$.

Substituting (15) into (14) results in

$$\begin{aligned} \dot{V}_{i,1} \leq& -c_{i,1}e_{i,1}^{p_i+1} + 2e_{i,1}^{p_i+1} + e_{i,2}^{p_i-p_{i,2}+1}\bar{\pi}_{i,1} - \frac{\mu_{i,1}}{2\tau_{i,1}}\tilde{\theta}_{i,1}^2 \\ &+ \gamma_{i,1}^{-\frac{p_i+1}{p_{i,1}}} + \eta_{i,1}^{-\frac{p_i+1}{p_{i,1}}}\xi_{i,1}^{\frac{p_i+1}{p_{i,1}}} + \frac{\mu_{i,1}}{2\tau_{i,1}}\theta_{i,1}^2 \end{aligned} \tag{16}$$

Step k ($k = 2, \ldots, n_i$): Similar to step 1, consider a Lyapunov function candidate as

$$V_{i,k} = \frac{e_{i,k}^{p_i-p_{i,k}+2}}{p_i-p_{i,k}+2} + \frac{1}{2l_{i,k}}\tilde{\theta}_{i,k}^2 \tag{17}$$

Construct the virtual control laws as

$$\alpha_{i,k} = -e_{i,k}\pi_{i,k} \tag{18}$$

where $\pi_{i,k} = \{\frac{1}{g_{\min}}(c_{i,k} + \gamma_{i,k}^{\frac{p_i+1}{p_i-p_{i,k}+1}}\hat{\theta}_{i,k}(S_{i,k}^T S_{i,k})^{\frac{p_i+1}{2(p_i-p_{i,k}+1)}} + \eta_{i,k}^{\frac{p_i+1}{p_i-p_{i,k}+1}})\}^{\frac{1}{p_{i,k}}}$, $c_{i,k} > 0$ is the design parameter. The adaptive law is chosen as

$$\dot{\hat{\theta}}_{i,k} = l_{i,k}\gamma_{i,k}^{\frac{p_i+1}{p_i-p_{i,k}+1}}\left(S_{i,k}^T S_{i,k}\right)^{\frac{p_i+1}{2(p_i-p_{i,k}+1)}} e_{i,k}^{p_i+1} - \mu_{i,k}\hat{\theta}_{i,k} \tag{19}$$

where $l_{i,k}$ and $\mu_{i,k}$ are the positive design parameters.

Then, we obtain

$$\begin{aligned} \dot{V}_{i,k} \leq& -\left(c_{i,k}-1\right)e_{i,k}^{p_i+1} + e_{i,k+1}^{p_i-p_{i,k+1}+1}\bar{\pi}_{i,k} - \frac{\mu_{i,k}}{2\tau_{i,k}}\tilde{\theta}_{i,k}^2 \\ &+ \gamma_{i,k}^{-\frac{p_i+1}{p_{i,k}}} + \eta_{i,k}^{-\frac{p_i+1}{p_{i,k}}}\xi_{i,k}^{\frac{p_i+1}{p_{i,k}}} + \frac{\mu_{i,k}}{2\tau_{i,k}}\theta_{i,k}^2 \end{aligned} \tag{20}$$

where $\bar{\pi}_{i,k} = e_{i,k+1}^{p_{i,k+1}}((2^{p_{i,k}-2}g_{\max}p_{i,k})^{p_i+1/p_{i,k}} + (2^{p_{i,k}-2}p_{i,k}\pi_{i,k}^{p_{i,k}-1})^{p_i+1} + (g_{\max}p_{i,k}\pi_{i,k}^{p_{i,k}-1})^{p_i+1})$.

At the present stage, the theorem can be concluded as follows.

Theorem 1: Consider the leader-following nonlinear MASs (3). If Assumption 1 holds, if we designed the virtual control as in (12), (18) with the adaptive laws (13), (19) ensures that all the agents achieve the bipartite consensus and the bipartite tracking error can converge to a small neighborhood of origin.

Proof: Construct the Lyapunov function

$$V = \sum_{i=1}^{N}\sum_{k=1}^{n_i} V_{i,k} \tag{21}$$

Obviously, if we choose $c_{i,1} > 2$, $c_{i,k} > 1$ ($2 \leq k \leq n_i - 1$) and $c_{i,n_i} > 0$, and $a_{i,0} = \min_{2\leq k\leq n_i-1}\{c_{i,1}-2, c_{i,k}-1, c_{i,n_i}\}$, $b_i = \sum_{k=1}^{n_i}(\gamma_{i,k}^{-\frac{p_i+1}{p_{i,k}}} + \eta_{i,k}^{-\frac{p_i+1}{p_{i,k}}}\xi_{i,k}^{\frac{p_i+1}{p_{i,k}}} + \frac{\mu_{i,k}}{2\tau_{i,k}}\theta_{i,k}^2)$. Then, we have

$$\begin{aligned} \dot{V} &\leq \sum_{i=1}^{N}\left\{-a_{i,0}\sum_{k=1}^{n_i} e_{i,k}^{p_i+1} - \sum_{k=1}^{n_i}\frac{\mu_{i,k}}{2\tau_{i,k}}\tilde{\theta}_{i,k}^2 + b_i\right\} \\ &\leq \sum_{i=1}^{N}\left\{-a_{i,0}(p_i+1)\sum_{k=1}^{n_i}\frac{e_{i,k}^{p_i-p_{i,k}+2}}{p_i-p_{i,k}+2} - \sum_{k=1}^{n_i}\frac{\mu_{i,k}}{2\tau_{i,k}}\tilde{\theta}_{i,k}^2\right. \\ &\quad \left. + \left(b_i + a_{i,0}\sum_{k=1}^{n_i}\frac{p_{i,k}-1}{p_i-p_{i,k}+2}\right)\right\} \\ &\leq \sum_{i=1}^{N}\left(-a_i V_{i,k} + \eta_i\right) \\ &\leq -aV + \eta \end{aligned} \tag{22}$$

where $a = \min\{a_i, i = 1, 2, \ldots, N\}$, $\eta = \sum_{i=1}^{N}\eta_i$, $a_i = \min_{1\leq k\leq n_i}\{a_{i,0}(p_i+1), \mu_{i,k}\}$, and $\eta_i = b_i + a_{i,0}\sum_{k=1}^{n_i}\frac{p_{i,k}-1}{p_i-p_{i,k}+2}$.

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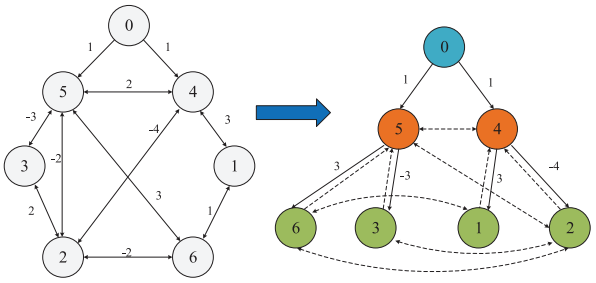


Fig. 1. Communications structurally balanced topology.

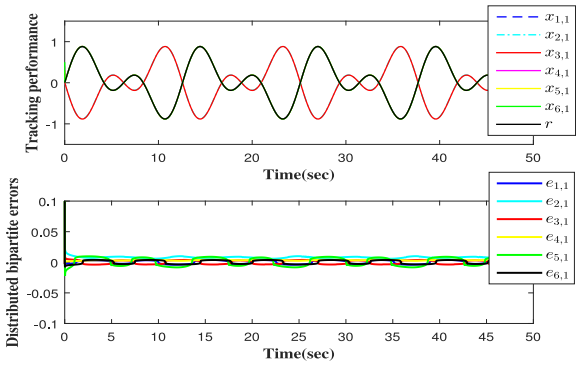


Fig. 2. The responses of tracking performance and tracking errors.

Furthermore, (22) satisfies,

$$0 \le V \le \left(V(0) - \frac{\eta}{a}\right)e^{-at} + \frac{\eta}{a}, t \ge 0. \tag{23}$$

Thus, according to (23), it can be obtained that all the signals in the MASs are bounded. Thus, the proof is finished. ■

IV. Simulation Studies

Consider the following second order nonlinear multi-agent systems:

$$\begin{cases} \dot{x}_{i,1} = g_{i,1}(\bar{x}_i)x_{i,2}^{p_{i,1}} + f_{i,1}(\bar{x}_i) \\ \dot{x}_{i,2} = g_{i,2}(\bar{x}_i)u_i^{p_{i,2}} + f_{i,2}(\bar{x}_i) \\ y_i = x_{i,1} \end{cases} \tag{24}$$

The nonlinear functions for the agents are chosen as $f_{1,1} = \sin(x_{1,1})$, $g_{1,1} = 1$, $p_{1,1} = 3$, $f_{1,2} = 2\cos(0.15x_{1,1}x_{1,2})$, $g_{1,2} = 1$, $p_{1,2} = 3$, $f_{2,1} = \cos(0.8x_{2,1})$, $g_{2,1} = 1 + x_{2,1}^2 + x_{2,2}^2$, $p_{2,1} = 3$, $f_{2,2} = 2\sin(0.15x_{2,1}x_{2,2})$, $g_{2,2} = 1 + x_{2,1}^2 + x_{2,2}^2$, $p_{2,2} = 5$, $f_{3,1} = 0$, $g_{3,1} = 1$, $p_{3,1} = 3$, $f_{3,2} = 2\sin(0.25x_{3,1}x_{3,2})$, $g_{3,2} = 1 + \sin^2(x_{3,1})$, $p_{3,2} = 5$, $f_{4,1} = \cos(x_{4,1})$, $g_{4,1} = 1$, $p_{4,1} = 3$, $f_{4,2} = 2\sin(0.15x_{4,1}x_{4,2})$, $g_{4,2} = 1$, $p_{4,2} = 3$. $f_{5,1} = \sin(0.8x_{5,1})$, $g_{5,1} = 1 + x_{5,1}^2 + x_{5,2}^2$, $p_{5,1} = 3$, $f_{5,2} = 2\cos(0.15x_{5,1}x_{5,2})$, $g_{5,2} = 1 + x_{5,1}^2 + x_{5,2}^2$, $p_{5,2} = 5$. $f_{6,1} = 0$, $g_{6,1} = 1$, $p_{6,1} = 3$, $f_{6,2} = 2\cos(0.25x_{6,1}x_{6,2})$, $g_{6,2} = 1 + \cos^2(x_{6,1})$, $p_{6,2} = 5$. In simulation, the initial conditions are given by $x_1(0) = [0.3, 0.2]^T$, $x_2(0) = [0.1, 0.2]^T$, $x_3(0) = [0.3, 0.4]^T$, $x_4(0) = [0.4, 0.1]^T$, $x_5(0) = [0.3, 0.2]^T$, $x_6(0) = [0.5, 0.2]^T$, $\hat{\theta}_{i,k}(0) = 0$ for all agents. The design parameters are chosen as $\mu_{i,k} = 0.2$, $l_{i,k} = 2$, $\eta_{i,k} = 5$, $\gamma_{i,1} = 100$, $\gamma_{i,2} = 10$, $c_{i,1} = 50$, $c_{i,2} = 20$ for $i = 1, \ldots, 6$ and $k = 1, 2$.

Case 1: Consider a balanced directed signed graph in Fig. 1. The leader's output is selected as $r(t) = 0.5(\sin(t)+\sin(0.5t))$.

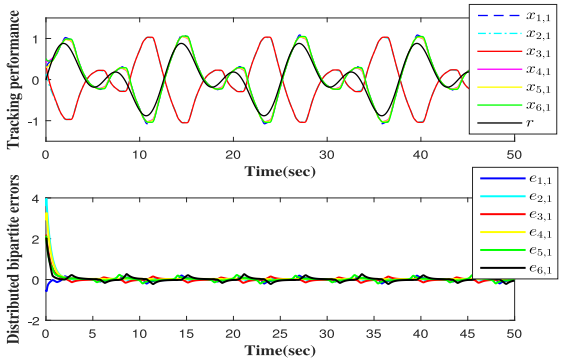


Fig. 3. The responses of tracking performance and tracking errors by using existing work [16].

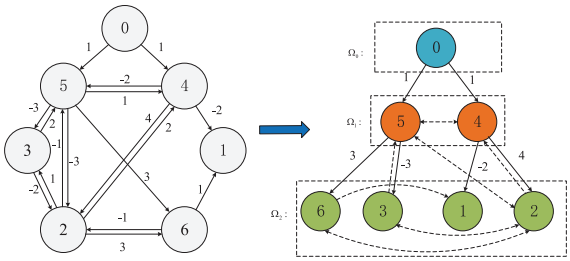


Fig. 4. Communications structurally unbalanced topology.

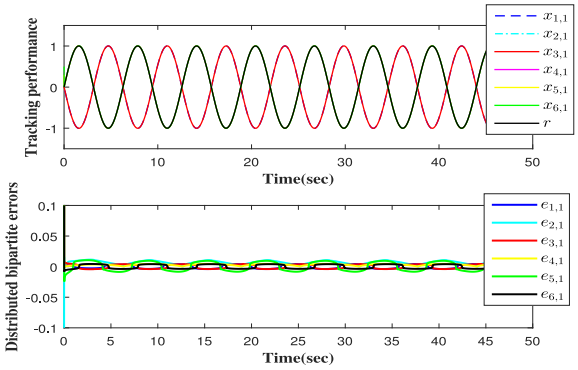


Fig. 5. The responses of tracking performance and tracking errors.

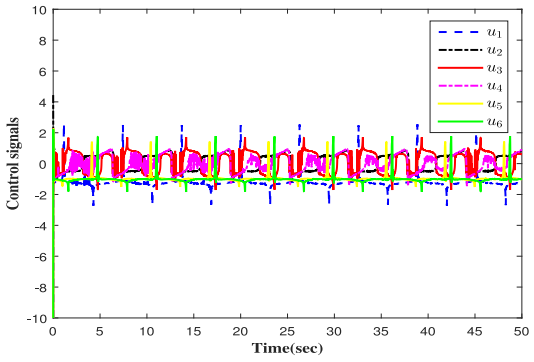


Fig. 6. The responses of control signals.

The simulation curves are shown in Figs. 2-3. Figs. 2-3 show that both our method and the existing results in the work [16] can achieve the bipartite consensuses. However, our proposed

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control strategy can guarantee a better bipartite consensus tracking performance than that in the work of [16].

Case 2: Consider an unbalanced directed signed graph in Fig. 4. The leader's output is selected as $r(t) = \sin(t)$. Here, the conventional gauge transformation cannot employed to the considered structurally unbalanced topology. However, based on a prioritized strategy, the agents in Fig. 4 can achieve a bipartition. Fig. 4 is given to show the evolution of connected structurally unbalanced topology by using prioritized strategy. Then, two different sets can be obtained by using prioritized strategy, i.e., $O_1 = \{o_0, o_2, o_4, o_5, o_6\}$ and $O_2 = \{o_1, o_3\}$. The simulation curves are shown in Figs. 5 - 6.

V. Conclusion

This brief has proposed the adaptive bipartite tracking control strategy for nonlinear MASs over signed directed graphs. By proposing a prioritized strategy, a more relaxation condition that a signed digraph only containing a spanning tree is used in this brief, and all the agents in the system can be achieved a bipartition. Under the proposed protocol, all the agents achieved the bipartite consensus with bounded errors. The feasibility of the newly designed method are verified by MATLAB simulation analysis. In the future work, we will pay attention to more complex MASs considering cyber attacks under structurally unbalanced and switching graphs.

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